

DESIGNERDATA

Thermo plastics

PS

Amorf. thermoplastic with low humid absorption; when foamed well known as EPS ('Expanded Polystyrene'); rarely re-enforced with glas fibers.

Specific properties

Mass: 1,05 g/cm³

Mechanical properties:

Stiff, hard, brittle. Low Charpy and impact resistance.

Thermal properties:

Applicable up to +70°C.

Burns very well with sooty flame without dripping.

Colour/surface

Glass clear transparent.

Can be coloured in almost any colour be it transparent or opaque; with high surface gloss.

Yellowing occur in UV-light

Processing

Well injection mouldable;

Well gluable with solvent glues (mostly based on Toluol, DiChlormethane, Butylacetate in which 2% PS is solved) ;

Gluable to other materials by 2-k glues.

Good weldable through different weld techniques;

Can be mechanically processed (milling, turning, drilling...)

Printable (tampon, film, silk screen printing)

Metallising through vacuum

Applications

- High gloss transparent packaging for cosmetics, pen's, cd's and dvd's, d-i-y articles, desk accessories ...

- Lights in all sorts for interior lighting ...

- displays in electronics ...

- Simple toys, model kits ...,

- disposable dinnerware (aerospace), storage aids, combs, toothbrushes, ...

Polystyrene

<u>Young's modulus</u>	3250	MPa
Shear modulus	750	MPa
<u>Tensile strength</u>	46	MPa
<u>Elongation</u>	3.4	%
<u>Fatigue</u>	20	MPa

Bending strength	75	MPa
Hardness	72.5	Rockwell
Impact strength	5	J/cm
Yield strength	0	MPa
Thermal expansion	70	E-6/K
Thermal conductivity	0.15	W/m*K
Specific heat	1300	J/kg*K
Vicat	100	°C
Melting temperature	Amorphou	°C
Glass temperature	89	°C
Minimum service temperature	-10	°C
Maximum service temperature	90	°C
Density	1045	kg/m ³
Resistivity	1E+22	Ohm*mm ² /m
Breakdown potential	130	kV/mm
Dielectric loss factor	0.0001	
Friction coefficient	0.4	
Refraction index	1.585	
Shrinkage	0.5	%
Water absorption	0.1	%